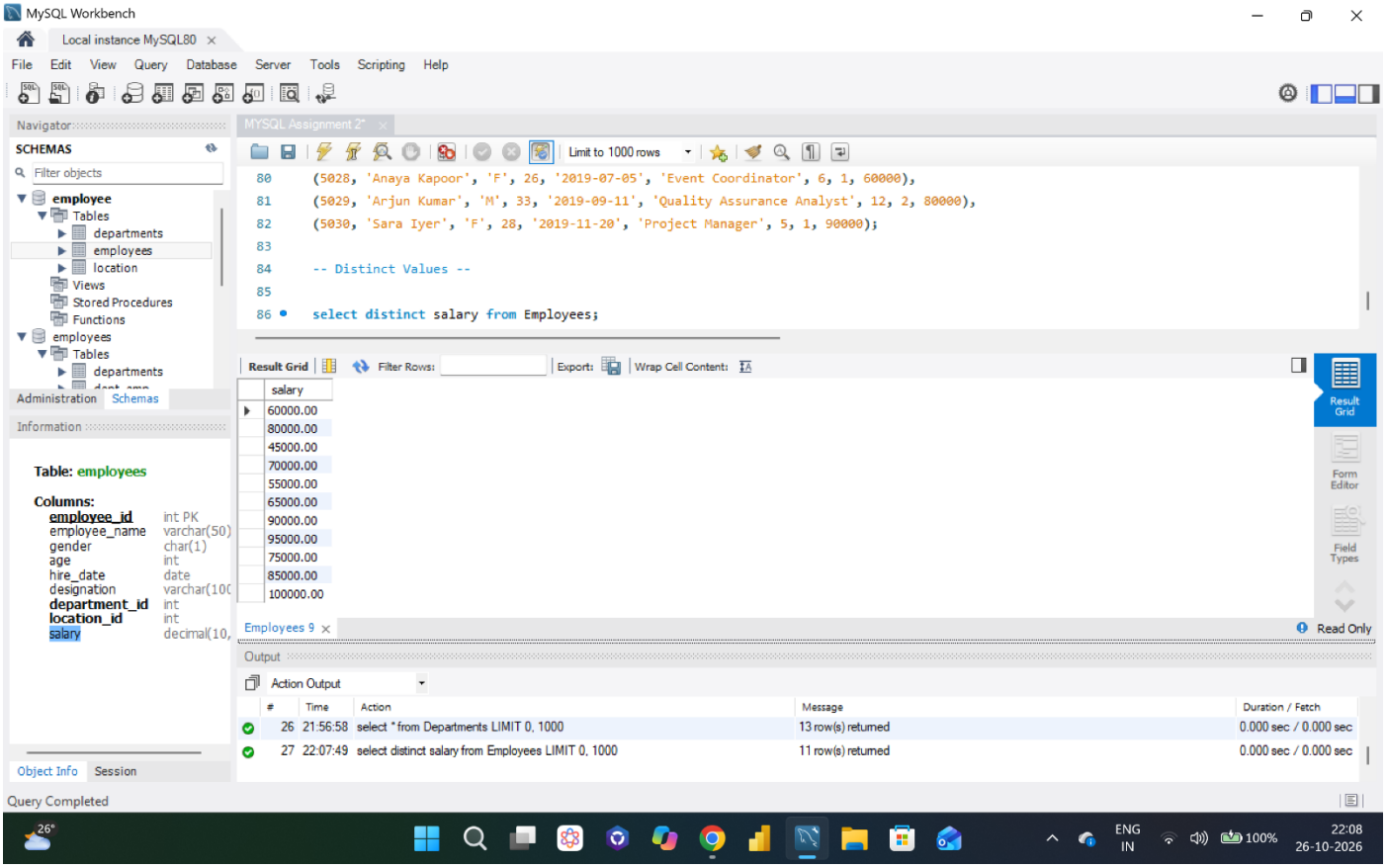
MYSQL ASSIGNMENT 2 – QUERYING DATA

- **Distinct Values:**

Retrieve distinct salaries from the Employees table.

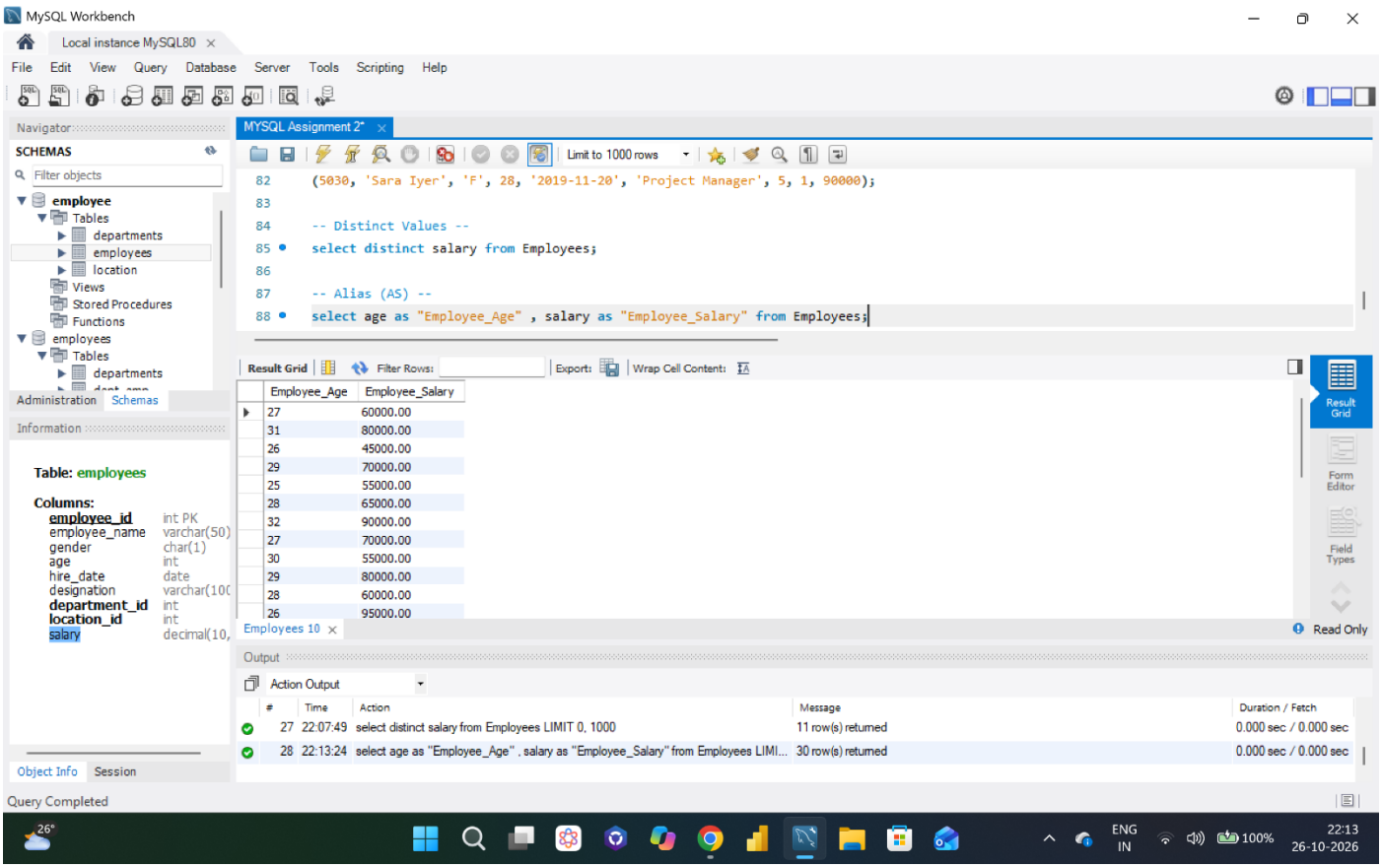
Query: select distinct salary from Employees;



- **Alias (AS):**

Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

Query: select age as "Employee_Age" , salary as "Employee_Salary" from Employees;

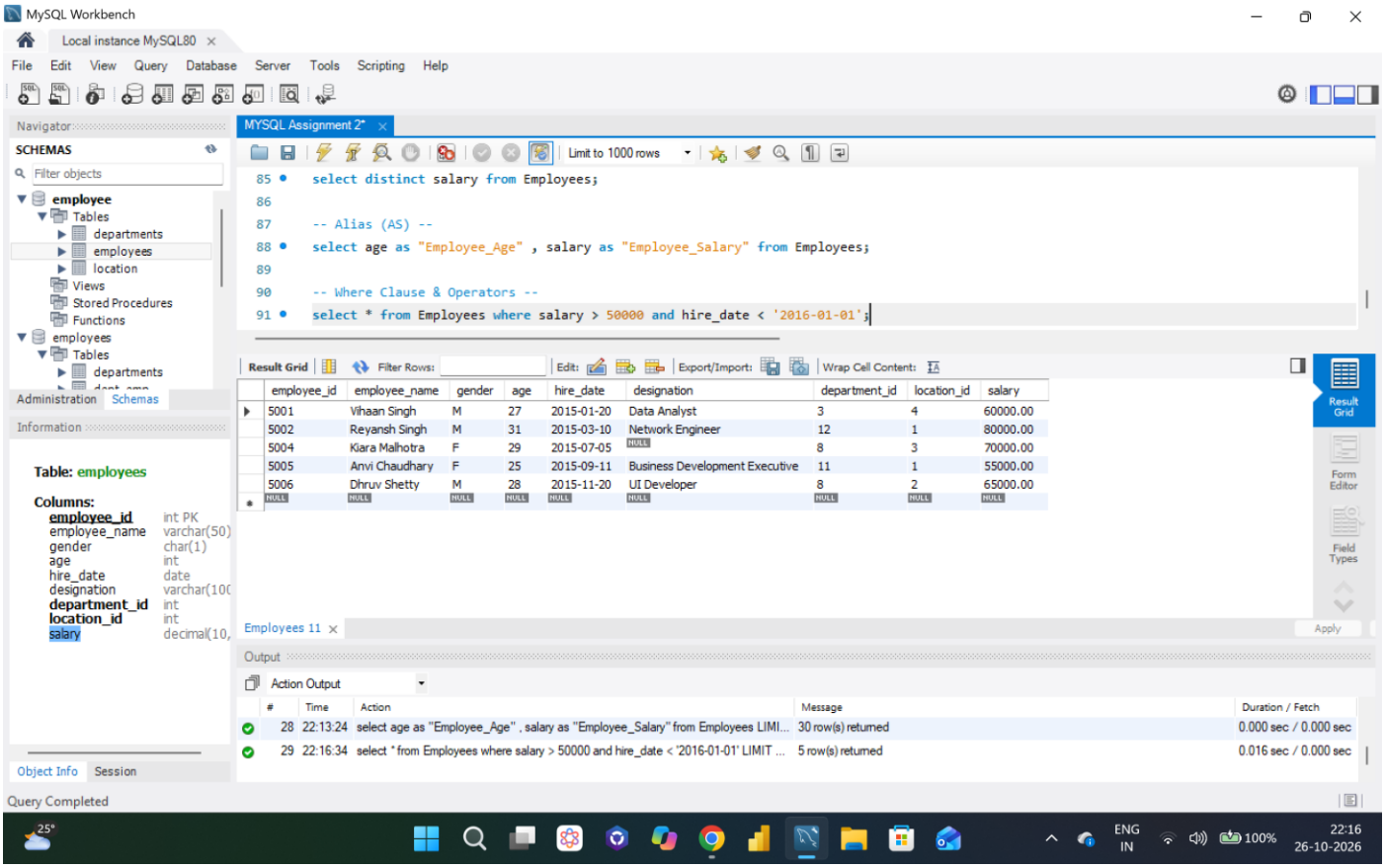


- **Where Clause & Operators:**

1. Retrieve employees with a salary greater than ₹50000 and hired before

2016-01-01.

Query: select * from Employees where salary > 50000 and hire_date < '2016-01-01';

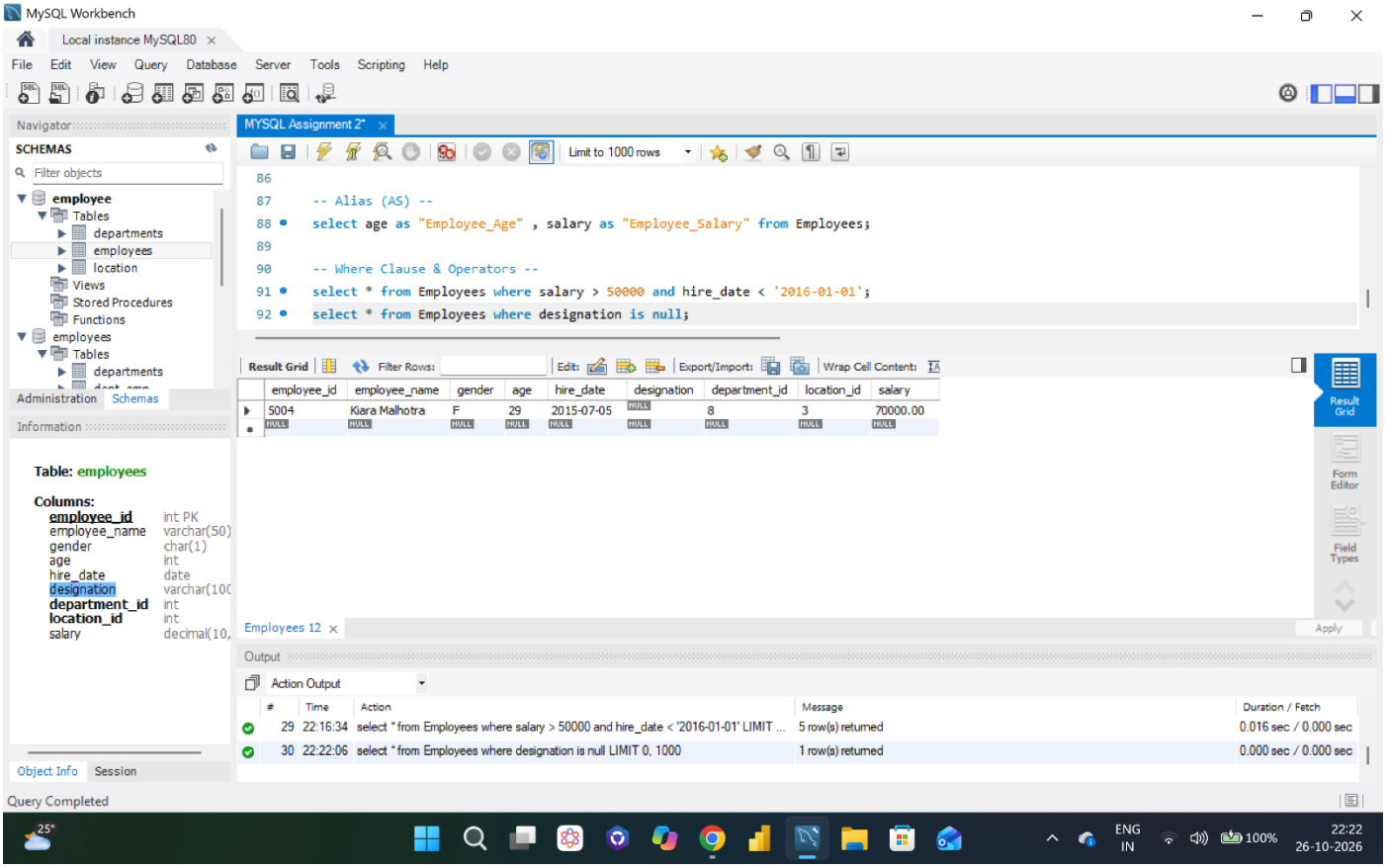


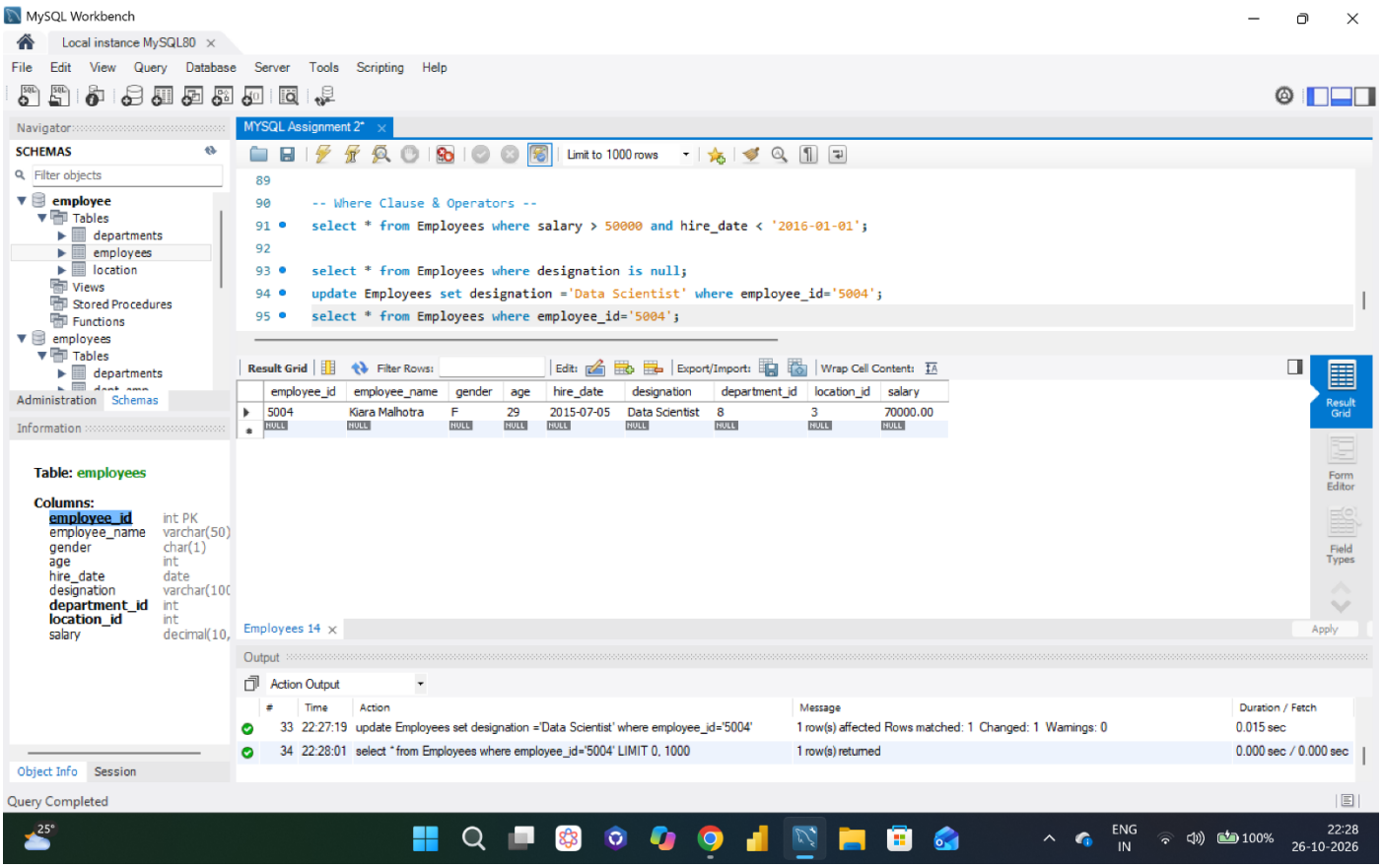
2. Find the employee whose designation is missing and fill it with "Data Scientist".

Query: select * from Employees where designation is null;

update Employees set designation ='Data Scientist' where employee_id='5004';

select * from Employees where employee_id='5004';



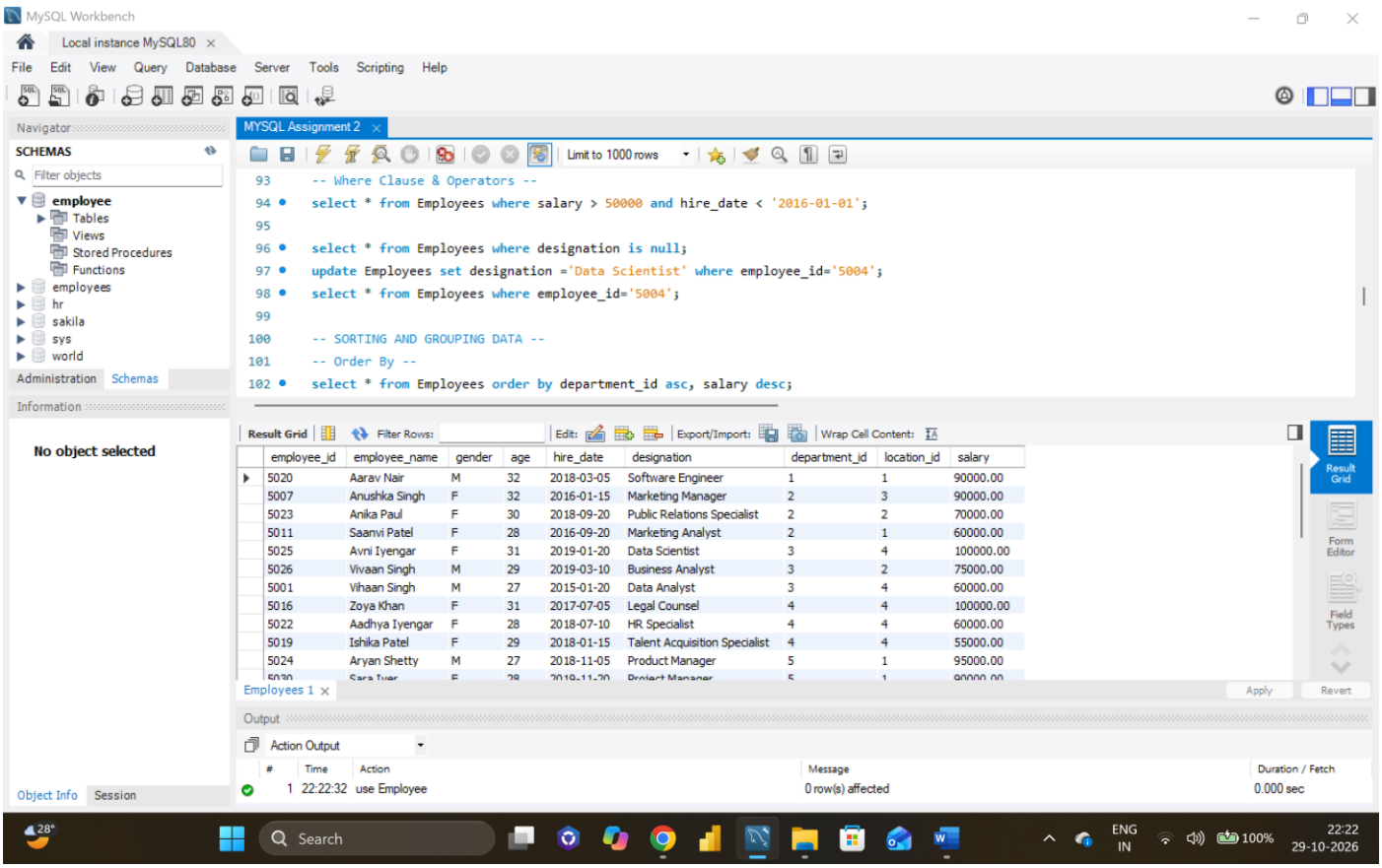


Sorting and Grouping Data:

1. Order by:

Find employees sorted by department ID in ascending order and salary in descending order.

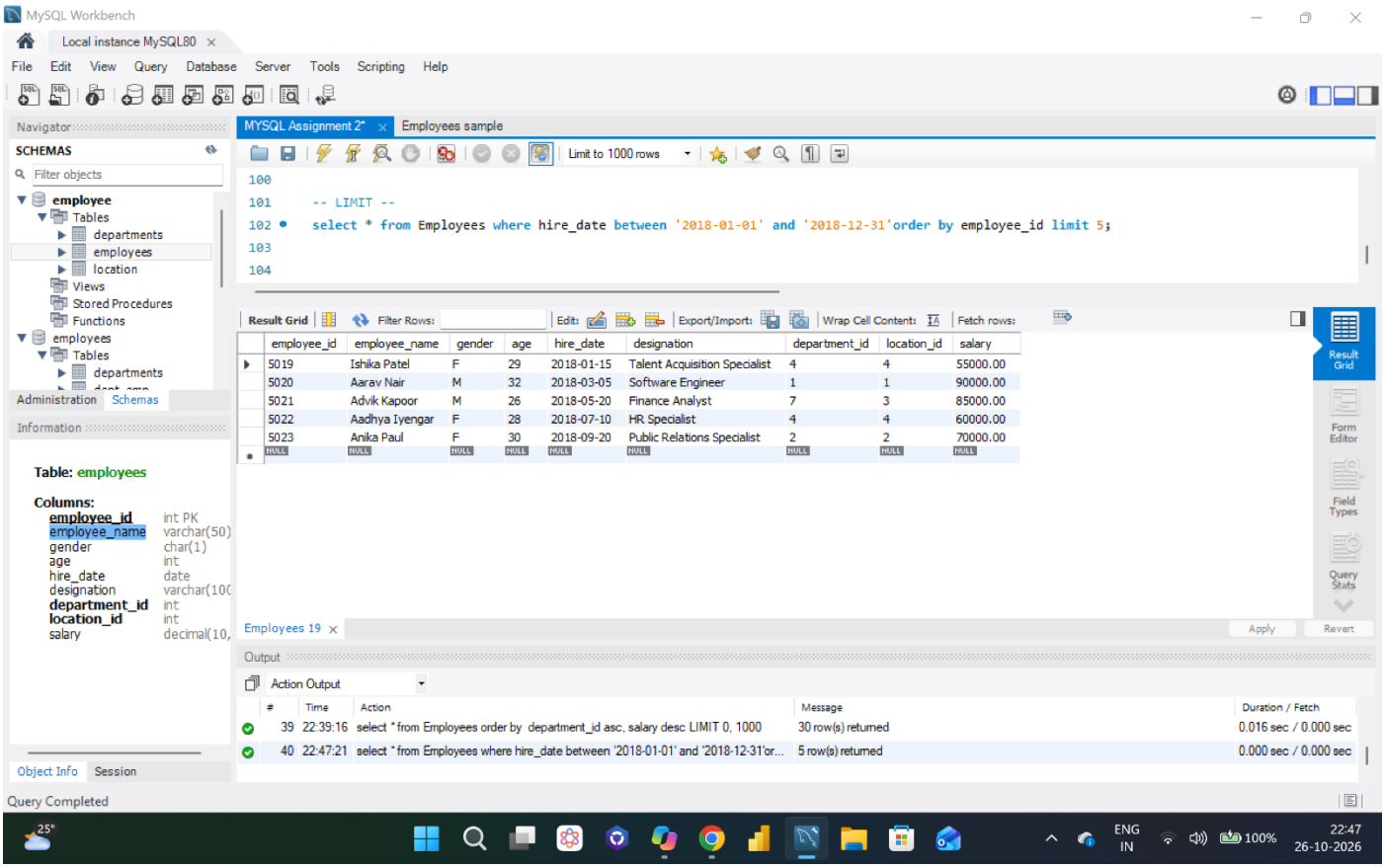
Query: select * from Employees order by department_id asc, salary desc;



2. Limit:

Display the first 5 employees hired in the year 2018.

Query: select * from Employees where hire_date between '2018-01-01' and '2018-12-31'order by employee_id limit 5;

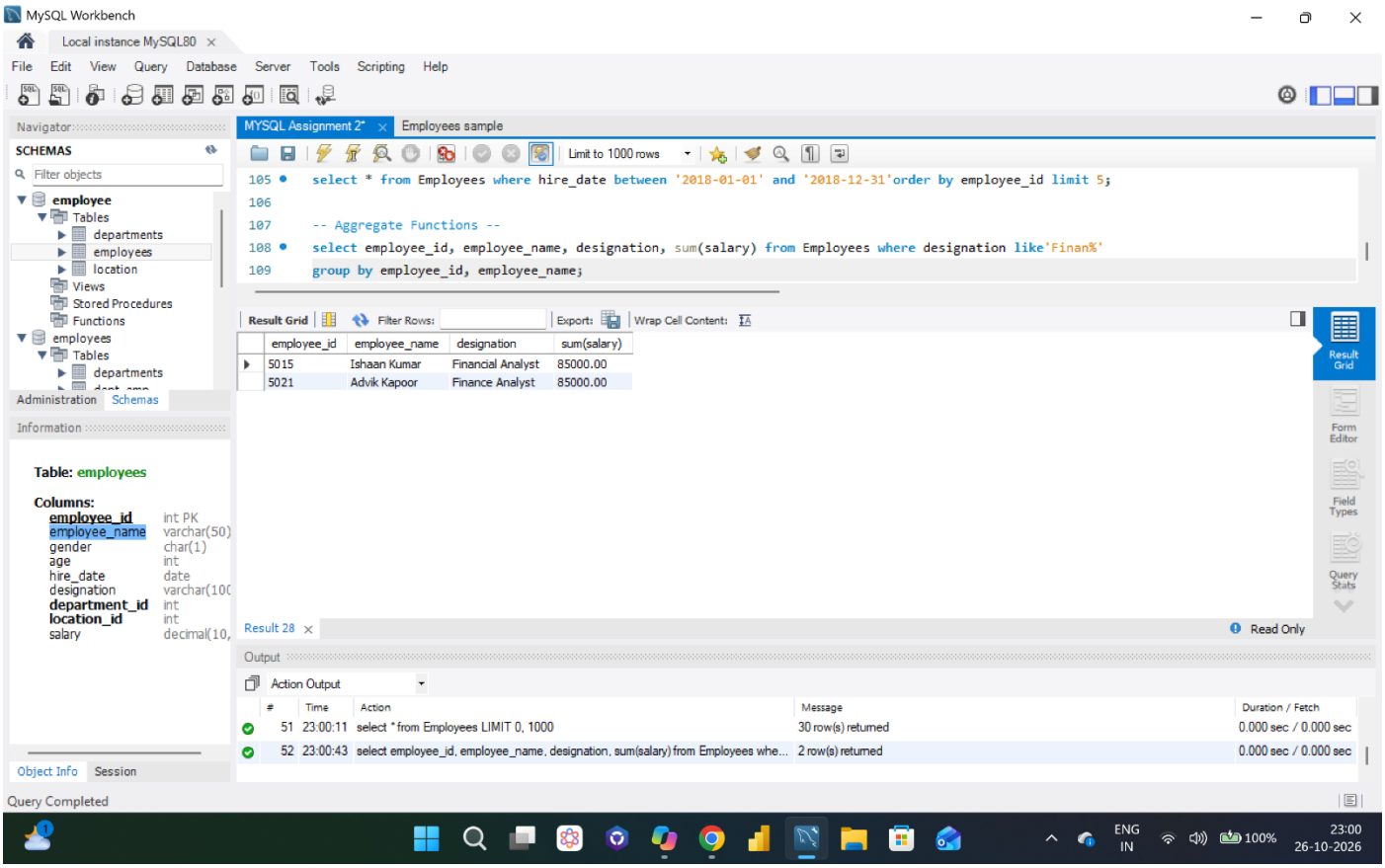


3. **Aggregate Functions:**

- Calculate the sum of all salaries in the Finance department.

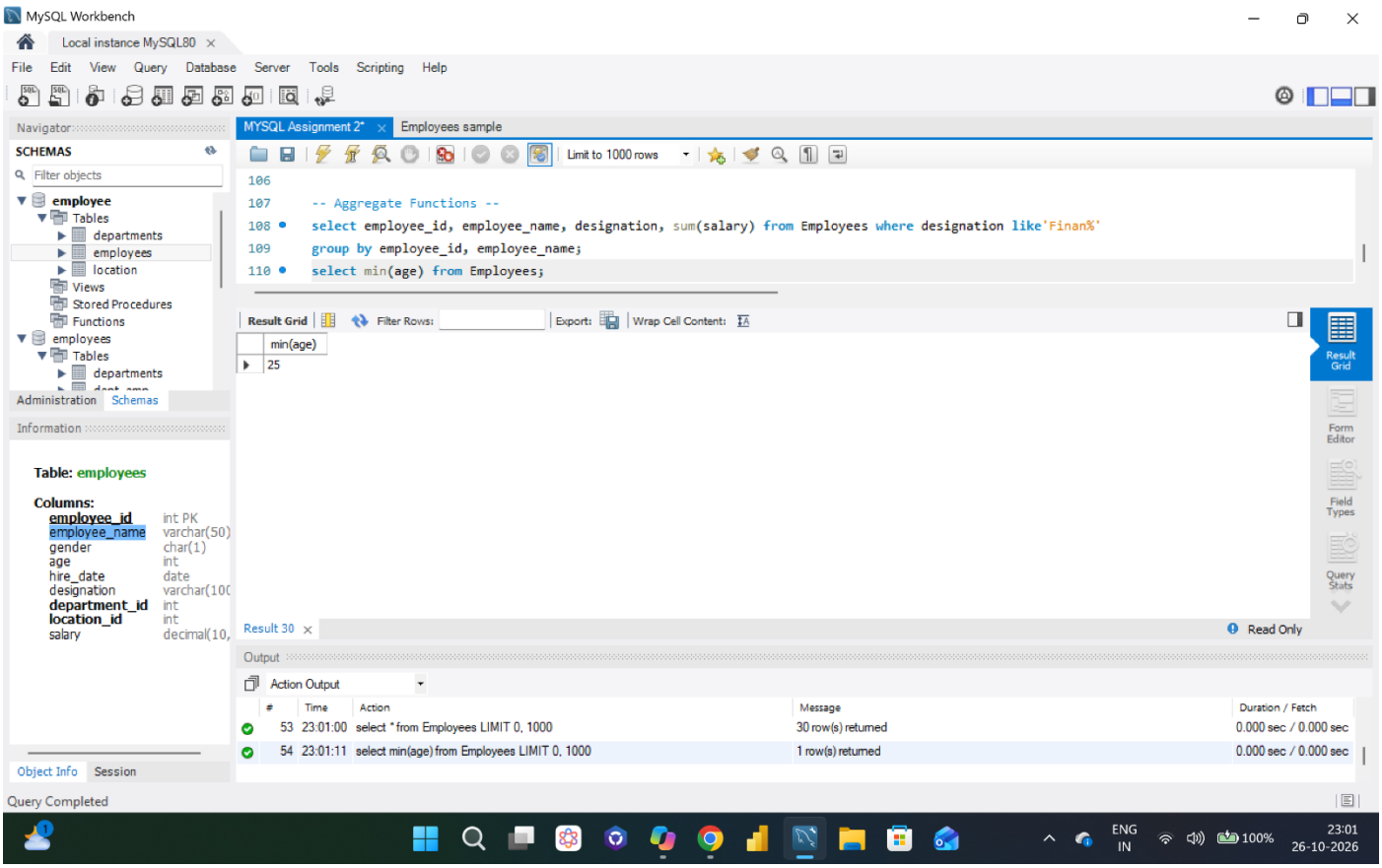
Query: select employee_id, employee_name, designation, sum(salary) from Employees where designation like'Finan%'

group by employee_id, employee_name;



- Find the minimum age among all employees.

Query: select min(age) from Employees;



4. **GROUP BY:**

- List the maximum salary for each location.

Query: select max(e.salary) Maximum_Salary, l.location

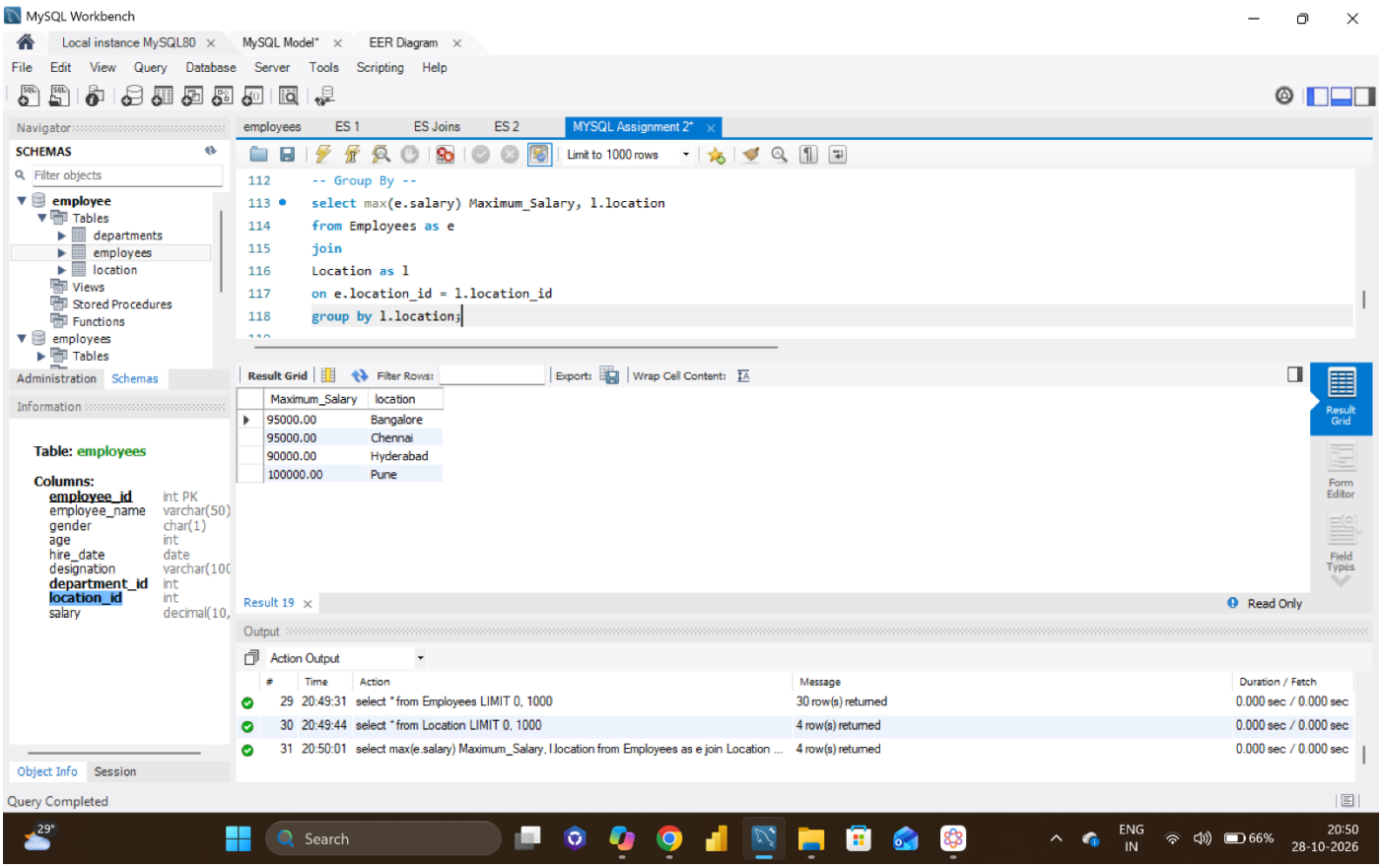
from Employees as e

join Location as l

on e.location_id = l.location_id

group by l.location

order by l.location;



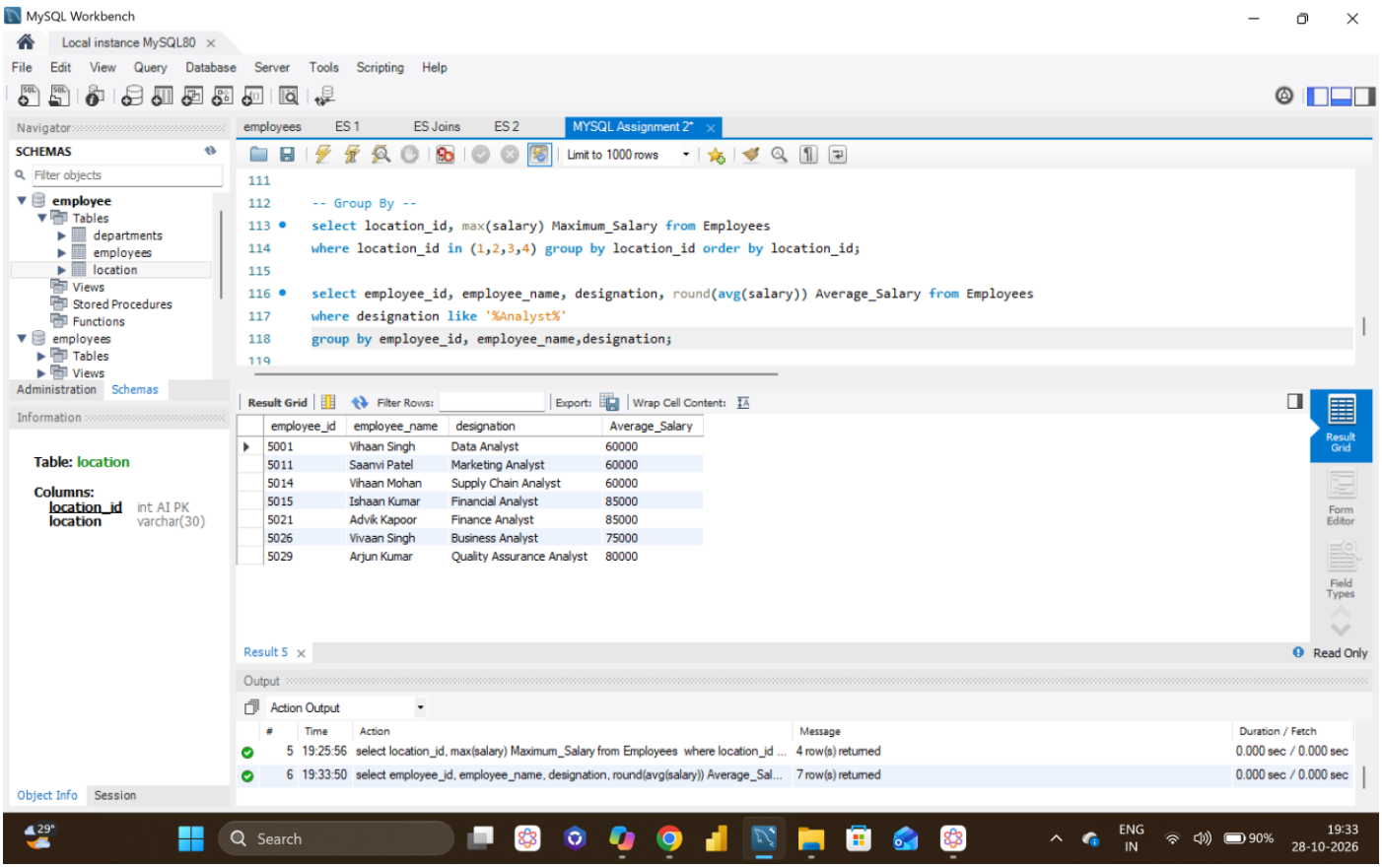
- Calculate the average salary for each designation containing the word

'Analyst'.

Query: select employee_id, employee_name, designation, round(avg(salary)) Average_Salary from Employees

where designation like '%Analyst%'

group by employee_id, employee_name,designation;



5. **HAVING**:

- Find departments with less than 3 employees.

Query: select d.department_name, count(e.employee_id) Count_of_Employees

from Employees as e

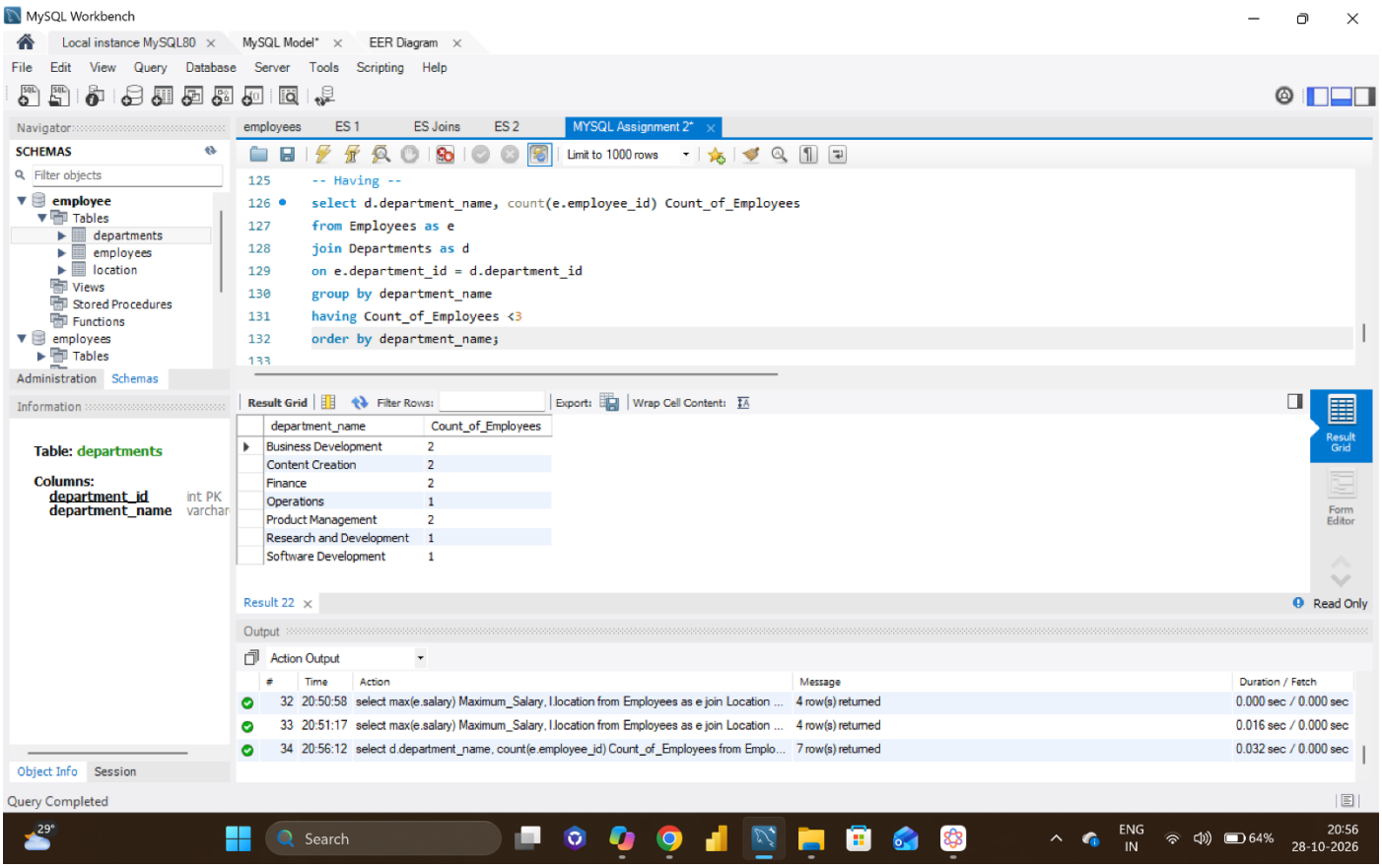
join Departments as d

on e.department_id = d.department_id

group by department_name

having Count_of_Employees <3

order by department_name;



- Find locations with female employees whose average age is below 30.

Query: select l.location, e.gender, avg(e.age) Average_Age

from Employees as e

join Location as l

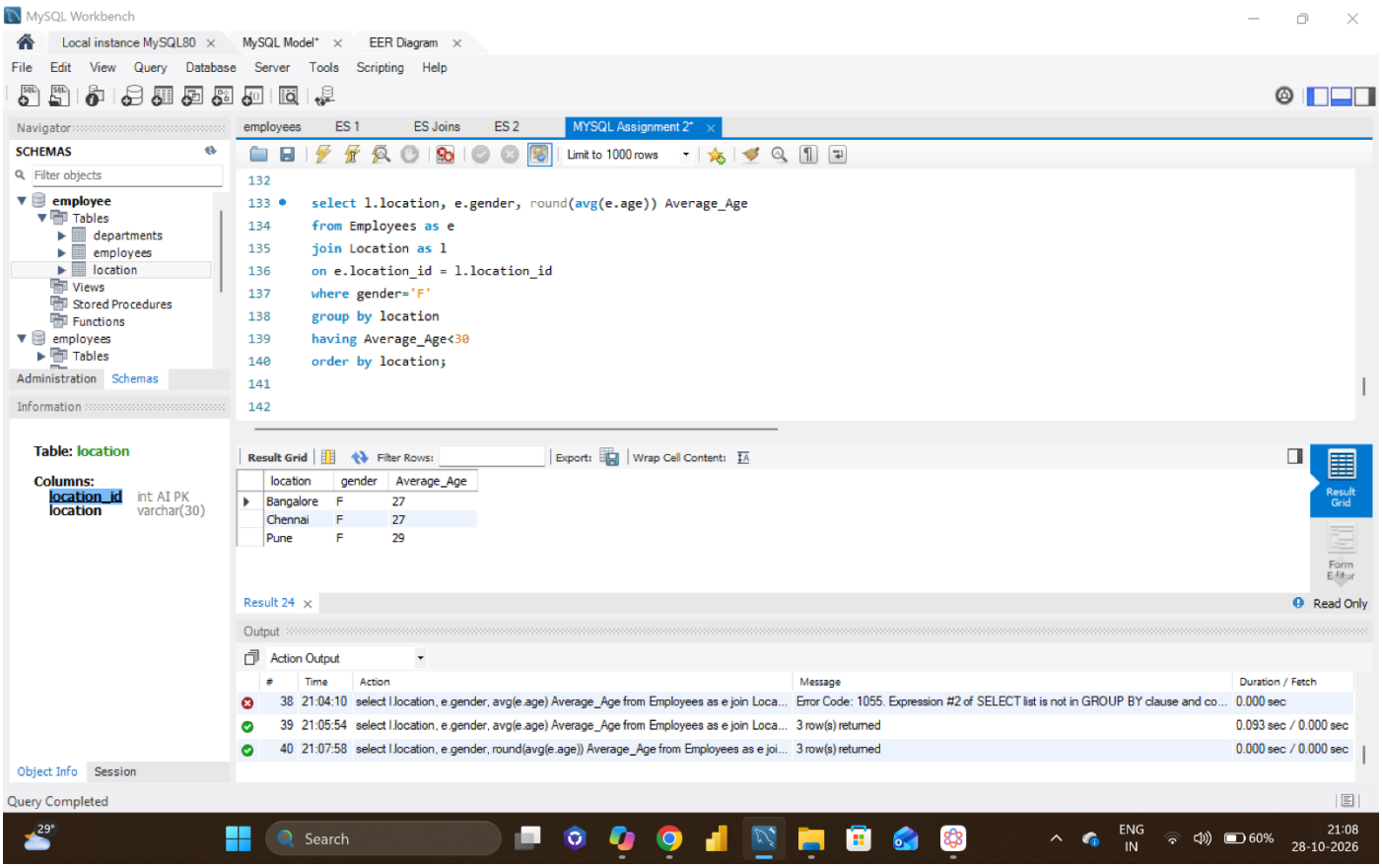
on e.location_id = l.location_id

where gender='F'

group by location

having Average_Age<30

order by location;



Joins:

1. **Inner Join:**

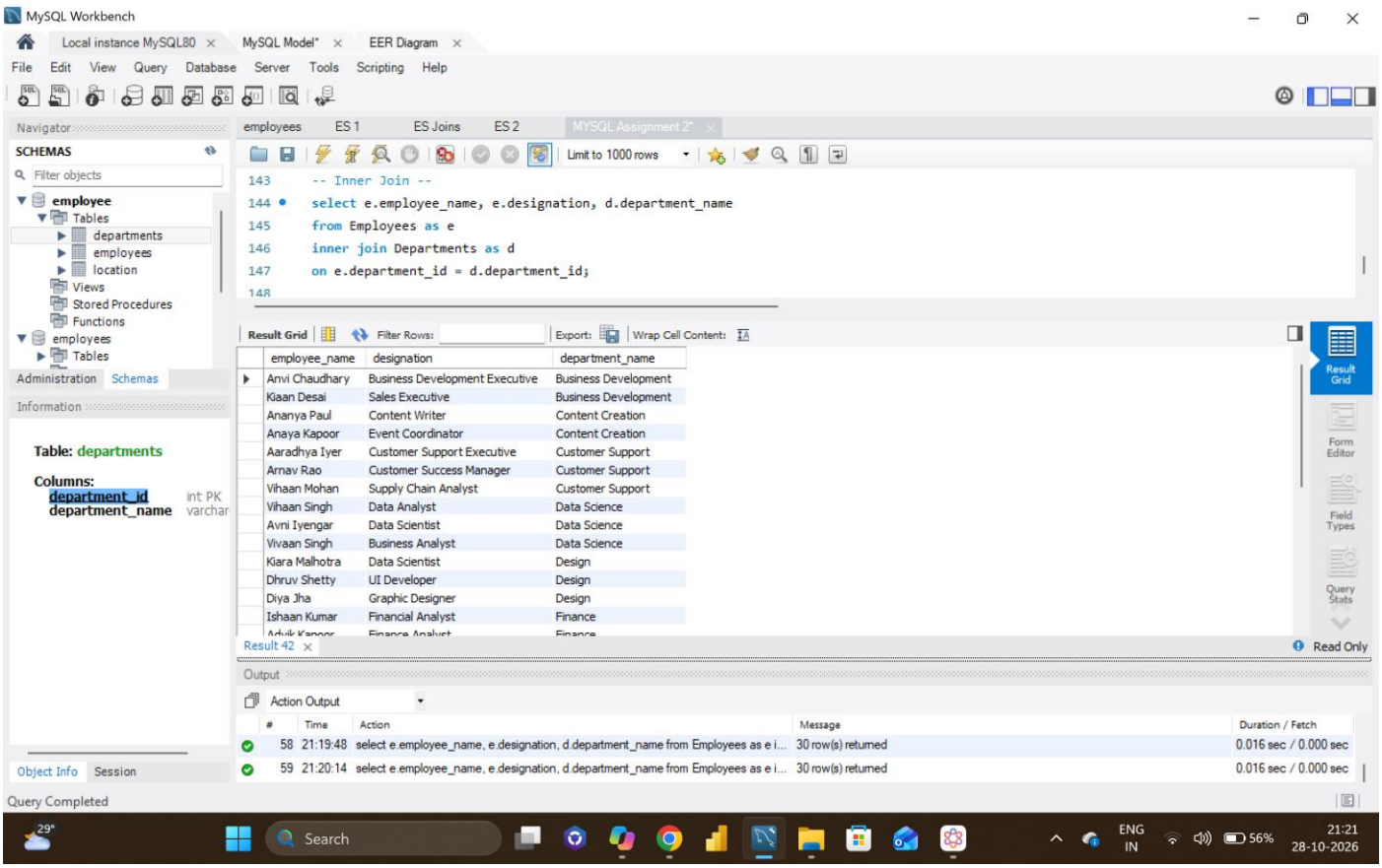
List employee names, their designations, and department names where employees are assigned to a department.

Query: select e.employee_name, e.designation, d.department_name

from Employees as e

inner join Departments as d

on e.department_id = d.department_id;



2. Left Join:

List all departments along with the total number of employees in each department, including departments with no employees.

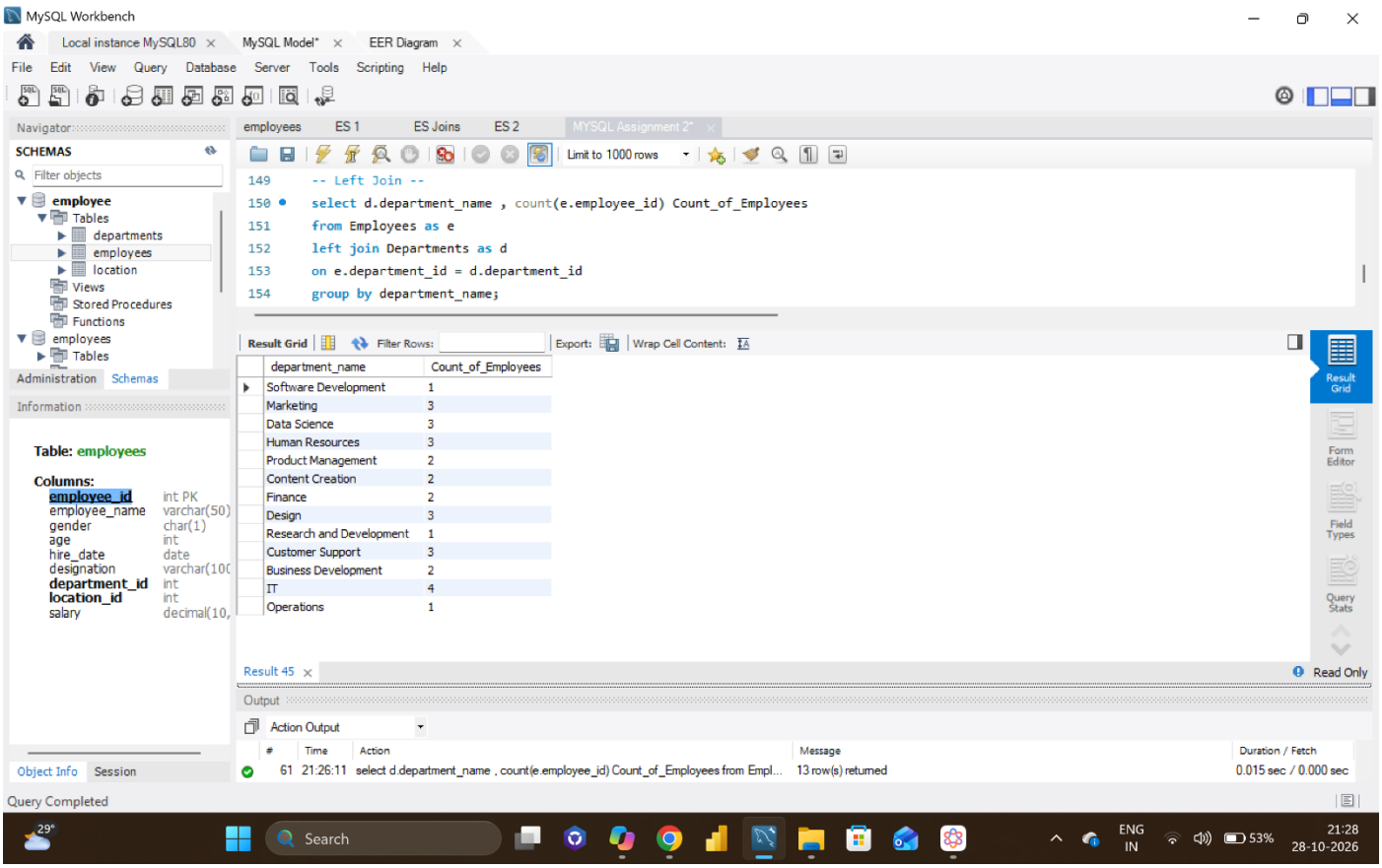
Query: select d.department_name , count(e.employee_id) Count_of_Employees

from Employees as e

left join Departments as d

on e.department_id = d.department_id

group by department_name;



3. Right Join:

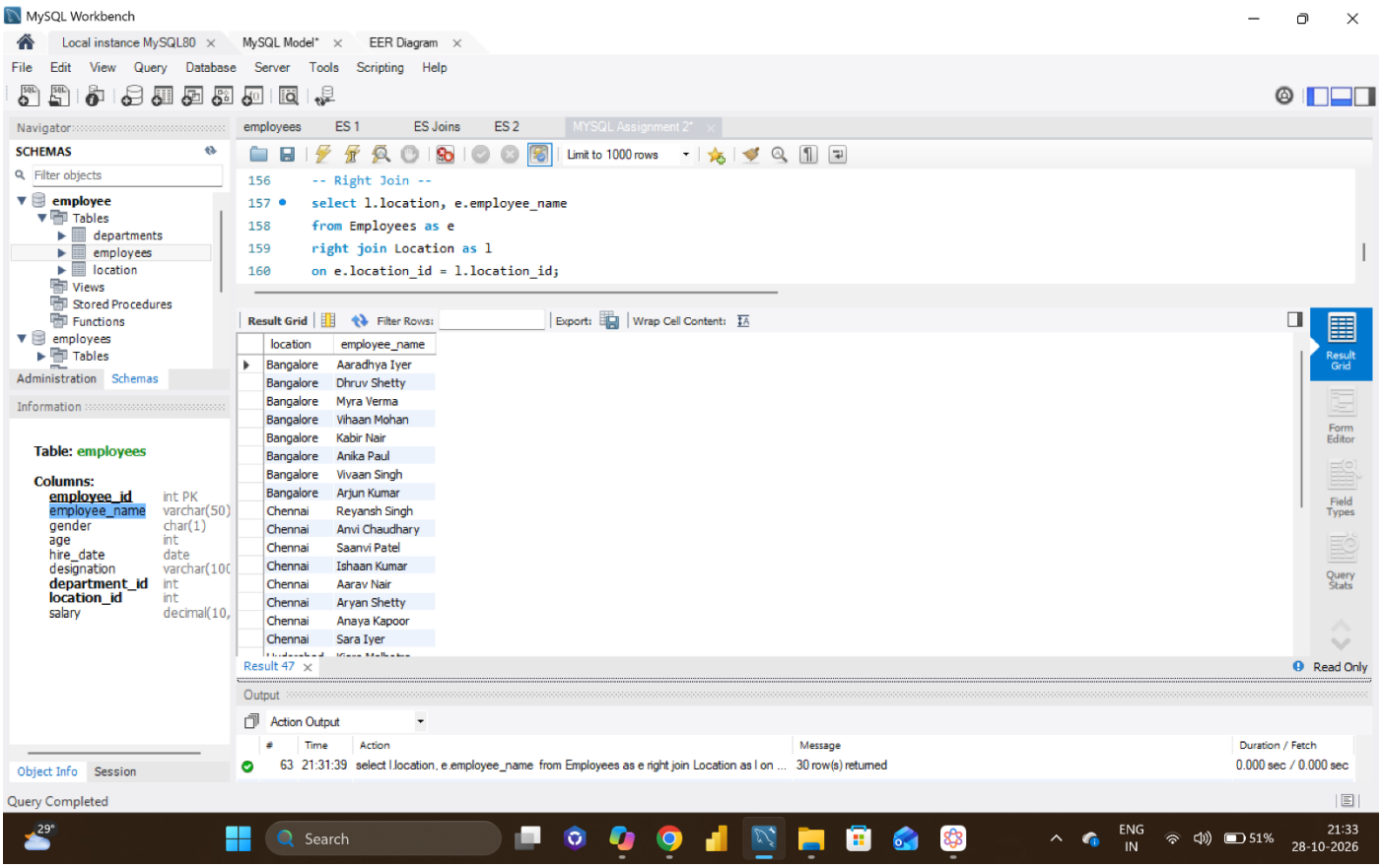
Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

Query: select l.location, e.employee_name

from Employees as e

right join Location as l

on e.location_id = l.location_id;


MySQL Workbench
Local instance MySQL80
MySQL Model*
EER Diagram
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
employee
Tables
departments
employees
location
Views
Stored Procedures
Functions
employees
Tables
Administration Schemas
Information
Table: employees
Columns:
employee_id int PK
employee_name varchar(50)
gender char(1)
age int
hire_date date
designation varchar(10(
department_id int
location_id int
salary decimal(10,
Object Info Session
employees ES 1 ES Joins ES 2 MYSQL Assignment 2*
Limit to 1000 rows
156 -- Right Join --
157 select l.location, e.employee_name
158 from Employees as e
159 right join Location as l
160 on e.location_id = l.location_id;
Result Grid Filter Rows: Export: Wrap Cell Content:
location employee_name
Bangalore Aaradhya Iyer
Bangalore Dhruv Shetty
Bangalore Myra Verma
Bangalore Vihaan Mohan
Bangalore Kabir Nair
Bangalore Anika Paul
Bangalore Vivaan Singh
Bangalore Arjun Kumar
Chennai Reyansh Singh
Chennai Anvi Chaudhary
Chennai Saanvi Patel
Chennai Ishaan Kumar
Chennai Aarav Nair
Chennai Aryan Shetty
Chennai Anaya Kapoor
Chennai Sara Iyer
Result Grid
Form Editor
Field Types
Query Stats
Result 47
Read Only
Output
Action Output
Time Action Message Duration / Fetch
63 21:31:39 select l.location, e.employee_name from Employees as e right join Location as l on ... 30 row(s) returned 0.000 sec / 0.000 sec
Query Completed
29°
Search
ENG IN
51%
21:33 28-10-2026